Mass Density Profiles in Globular and Open Clusters

A Comparative Study of M2 and MS/34

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Scientific Motivation

- Star clusters are natural laboratories for studying **stellar evolution** and **gravitational dynamics**. [Binney and Tremaine, 2008]
- Through countless two-body encounters, more massive stars lose kinetic energy and **sink toward the cluster center**, while lighter stars migrate outward.
- This process, known as **mass segregation**, reveals the **dynamical age** of a cluster — how far it has evolved toward energy equipartition.
- Comparing a **young open cluster (M34)** with an **old globular cluster (M2)** allows us to test how mass segregation strengthens with time and stellar density. [Harris, 2010, Shapiro et al., 2010]
- Understanding this helps explain how clusters **evolve, dissolve, and contribute stars to the Galactic field**.

Concept: Mass Segregation in Star Clusters

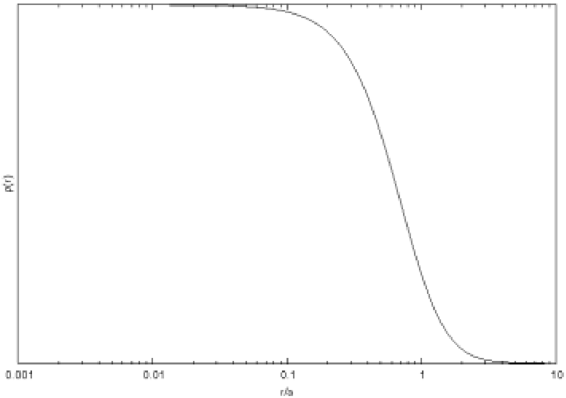
- Gravitational encounters redistribute energy between stars.
- **Massive stars** lose energy and migrate inward.
- **Low-mass stars** gain energy and drift outward.
- The effect becomes stronger as the cluster dynamically relaxes.
- Observable through radial distributions or CMD-based mass gradients.

Introduction to Star Clusters

- Gravitationally bound groups of stars that formed from the same giant molecular cloud.
- Act as a natural laboratory for studying stellar evolution.
- Two main types: Open clusters and Globular clusters.

What is a Mass Density Profile?

- A function that describes how the density of stars or mass changes with distance from a cluster's center.
- Provides insight into a cluster's internal structure and dynamical evolution.



The Plummer Model

- Proposed in 1911 as a mathematical model for globular clusters. [Plummer, 1911]
- Provides a simple, analytically tractable representation of a spherical system, featuring a core of constant density that transitions to a steeper fall-off at larger radii.
- This model is particularly useful for studying the dynamics of clusters, as it offers a smooth, continuous description of the mass distribution.

Plummer Model: The Potential $\Phi(r)$

- The model is defined by its gravitational potential, $\Phi(r)$, which is given by:

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + a^2}}$$

  where:
  - G is the gravitational constant.
  - M is the total mass of the cluster.
  - a is the Plummer radius, a scale parameter that sets the size of the cluster's core.
- This potential approaches that of a point mass $(-\frac{GM}{r})$ for $r \gg a$.

Plummer Model: Deriving the Density $\rho(r)$

- The relationship between potential $\Phi(r)$ and mass density $\rho(r)$ is given by Poisson's equation:

$$\nabla^2\Phi = 4\pi G\rho$$

- In spherical coordinates, Poisson's equation for a spherically symmetric system simplifies to:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\Phi}{dr}\right) = 4\pi G\rho(r)$$

Derivation Step 1: Radial Derivative

- We begin by taking the first radial derivative of the potential $\Phi(r)$:

$$\Phi(r) = -GM(r^2 + a^2)^{-1/2}$$

$$\frac{d\Phi}{dr} = -GM\left(-\frac{1}{2}\right)(r^2 + a^2)^{-3/2}(2r)$$

$$\frac{d\Phi}{dr} = \frac{GMr}{(r^2 + a^2)^{3/2}}$$

Derivation Step 2: Second Radial Derivative

- Next, we multiply by r^2 and take the derivative again:

$$r^2 \frac{d\Phi}{dr} = \frac{GMr^3}{(r^2 + a^2)^{3/2}}$$

- Now, differentiate this expression with respect to r:

$$\begin{aligned}
\frac{d}{dr}\left(r^2 \frac{d\Phi}{dr}\right) &= GM \frac{d}{dr}\left(\frac{r^3}{(r^2 + a^2)^{3/2}}\right) \\
&= GM \left[\frac{3r^2(r^2 + a^2)^{3/2} - r^3 \cdot \frac{3}{2}(r^2 + a^2)^{1/2}(2r)}{(r^2 + a^2)^3}\right] \\
&= GM \left[\frac{3r^2(r^2 + a^2) - 3r^4}{(r^2 + a^2)^{5/2}}\right] \\
&= GM \left[\frac{3r^4 + 3a^2r^2 - 3r^4}{(r^2 + a^2)^{5/2}}\right] \\
&= \frac{3GMa^2r^2}{(r^2 + a^2)^{5/2}}
\end{aligned}$$

Derivation Step 3: Solve for $\rho(r)$

- Substitute the result back into Poisson's equation:

$$\frac{1}{r^2}\left(\frac{3GMa^2r^2}{(r^2+a^2)^{5/2}}\right) = 4\pi G\rho(r)$$

- Simplify and solve for $\rho(r)$:

$$\frac{3GMa^2}{(r^2+a^2)^{5/2}} = 4\pi G\rho(r)$$

$$\rho(r) = \frac{3M}{4\pi a^3}\frac{1}{\left(1+\frac{r^2}{a^2}\right)^{5/2}}$$

Plummer Model: Key Features

- **Central Core:** As $r \to 0$, the density approaches a constant value:

$$\rho(0) = \frac{3M}{4\pi a^3}$$

- **Outer Profile:** As $r \to \infty$, the density falls off steeply as r^{-5}:

$$\rho(r) \propto r^{-5}$$

[King, 1966]

S/N Formula

- **Signal:**

$$S = FA_\epsilon \tau$$

- **Noise:**

$$N_T^2 = N_R^2 + N_S^2 + N_{DC}^2 + N_\beta^2$$

- **Signal to Noise:**

$$S/N = \frac{FA_\epsilon \tau}{\left(N_R^2 + \tau(FA_\epsilon + i_{DC} + F_\beta A_\epsilon \Omega)\right)^{1/2}}$$

[Lubin, 2016, Howell, 2006]

Implementation

Listing 1: S-N Calculation

```
target_flux = 3500 # photons/s/m^2 for Mag 16 star in M2
area_effective = 0.036
integration_time = 100
noise_readout = 1
dark_current = 0.0022
background_flux = 2.6*10**12 # photons/s/m^2/st
pixel_solid_angle = 1.28*10**-11 # steradians

def snr(target_flux, area_effective, integration_time,
        noise_readout, dark_current,
        background_flux, pixel_solid_angle):
    signal = target_flux * area_effective * integration_time
    noise_background = background_flux * area_effective * integration_time *
         pixel_solid_angle
    noise_dark = dark_current * integration_time
    source_noise = signal
    total_noise = (source_noise+noise_background+noise_dark+noise_readout**2)**0.5
    snr = signal / total_noise
    return snr
```

Multipixel S/N

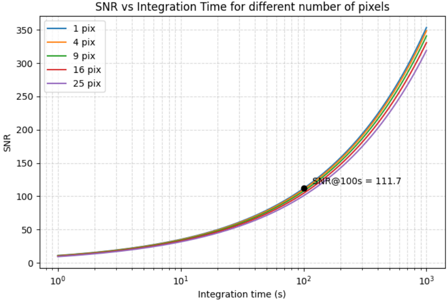


Figure: Multipixel S/N versus integration time

S/N Calculations

Object	Filter	Mag	# Exp	Int t	S/N	S/N stack
M2	SDSS g'	14	3	8	180–300	310–520
M2	SDSS g'	18	3	120	45–60	78–104
M2	SDSS g'	21.5	3	300	20–24	35–42
M34	SDSS g'	11	3	10	200–300	350–520
M34	SDSS g'	17.5	3	60	60–80	104–139
M34	SDSS g'	19	3	180	25–30	43–52
M2	SDSS r'	14	3	8	250–400	430–690
M2	SDSS r'	18	3	90	50–70	87–121
M2	SDSS r'	20.5	3	240	22–28	38–49
M34	SDSS r'	11	3	10	> 300	$\gg 500$
M34	SDSS r'	17.5	3	60	70–90	120–155
M34	SDSS r'	19	3	180	30–35	52–61

Table: S/N calculations for each filter

Image stacking to increase S/N ratio

Each exposure contains the same signal (star light) but different noise (random variations).
Adding or averaging multiple exposures reinforces the signal and averages down the noise.

$$S/N_{stacked} = S/N_{single} \times \sqrt{N}$$

Where N is the number of images.
This allows shorter individual exposures (avoids saturation) and improves photometric precision and faint-star detection.

Fundemental Problem

- **Completeness:**

$$C(m) := \frac{N_{detected}(m)}{N_{true}(m)}$$

- **Relation to LF:**

$$\Phi_{obs}(m) = C(m) \cdot \Phi_{true}(m)$$

Sources of Incompleteness

- Photon statistics
- Crowding
- Image quality

Mathematical Framework

$$P(m_{obs}|m_{true}) = \frac{1}{\sqrt{2\pi\sigma_m^2}} \exp\left[-\frac{(m_{obs} - m_{true})^2}{2\sigma_m^2}\right]$$

with $\sigma_m(m)$ as magnitude-dependent photometric uncertainty. So

$$C(m_{true}) = \int_{-\infty}^{m_{lim}} P(m_{obs}|m_{true}) dm_{obs}$$

with m_{lim} as limiting magnitude. So

$$C(m) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{m_{50} - m}{\sqrt{2}\sigma_{comp}}\right)\right]$$

with m_{50} as 50% completeness magnitude, σ_{comp} controls steepness

Position-Dependent Completeness

More realistically $C(m, \mathbf{r})$
Radius dependent crowding

$$C(m, \mathbf{r}) \approx C_0(m) \cdot f_{crowd}(\mathbf{r})$$

Background varies

$$C(m, x, y) \approx C_0(m) \cdot f_{bg}(x, y)$$

example

$$f_{crowd}(r) = \exp\left[-\alpha \frac{\Sigma(r)}{\Sigma_0}\right]$$

Methodology

- For a magnitude bin m_i in $[m_{min}, m_{max}]$ generate N_{add} stars with random positions (x_i, y_i) [Stetson, 1990]
- Use reduction pipeline
- Match recovered stars to input positions

$$\Delta r_{ij} = \sqrt{(x_i - x_j^{rec})^2 + (y_i - y_j^{rec})^2}$$

- say star recovered if

$$\Delta r < r_{match}, \qquad |m_{rec} - m_{input}| < \Delta m_{max}$$

Calculating Completeness

Using recovered stars

$$C(m_i) = \frac{N_{recovered}(m_i)}{N_{added}(m_i)}$$

and uncertainty

$$\sigma_C(m_i) = \sqrt{\frac{C(m_i)(1 - C(m_i))}{N_{added}(m_i)}}$$

Repeat artificial star test $N_{MC} \approx 50 - 100$ times

$$\sigma_C^{sys} = \mathrm{std}\left(\{C_i(m)\}_{i=1}^{N_{MC}}\right)$$

total uncertainty

$$\sigma_C^{total} = \sqrt{(\sigma_C^{stat})^2 + (\sigma_C^{sys})^2}$$

Fitting to Error Function Model

Using our set of completeness values $\{C_i \pm \sigma_i\}$ at magnitudes $\{m_i\}$ we fit and minimize

$$\chi^2(\theta) = \sum_i \frac{[C_i - C(m_i;\theta)]^2}{\sigma_i^2}$$

But least squares assumes errors are Gaussian (not true) and have constant variance (not true, depends on N). Instead, use binomial statistics. [Stetson, 1987, Bertin and Arnouts, 1996]

$$N_{rec} \approx \mathrm{Binomial}(N_{add}, C(m))$$

or

$$P(N_{rec}|N_{add}, C) = \binom{N_{add}}{N_{rec}} C^{N_{rec}} (1-C)^{N_{add}-N_{rec}} = \mathcal{L}(C|N_{rec}, N_{add})$$

Multiple Likelihood Bins

Say we test for magnitude bins $m_1, \ldots, m_n$ and we want to fit $C(m; \theta)$. Assuming independence of bins

$$\mathcal{L}(\theta|\text{data}) = \prod_{i=1}^{n} \binom{N_{add,i}}{N_{rec,i}} C(m_i; \theta)^{N_{rec,i}} [1 - C(m_i, \theta)]^{N_{add,i} - N_{rec,i}}$$

or

$$\ln \mathcal{L}(\theta) = \sum_{i=1}^{n} \{N_{rec,i} \ln C(m_i; \theta) + [N_{add,i} - N_{rec,i}] \ln[1 - C(m_i; \theta)]\}$$

and

$$\hat{\theta}_{MLE} = \arg\max_{\theta} \ln \mathcal{L}(\theta)$$

Fisher Information Matrix

Then for errors. Define the Fisher information matrix

$$F_{jk} = -\left\langle \frac{\partial^2 \ln \mathcal{L}}{\partial\theta_j \partial\theta_k} \right\rangle$$

and the parameter covariance matrix

$$\mathrm{Cov}(\theta) = F^{-1}$$

so $\sigma_{m_{50}} = \sqrt{[\mathrm{Cov}]_{11}}$ and $\sigma_{\sigma_{comp}} = \sqrt{[\mathrm{Cov}]_{22}}$

Model Comparison

Compare different completeness functions (error function vs. hyperbolic tangent vs. Fermi-Dirac)

$$\mathrm{AIC} = -2 \ln \mathcal{L}_{max} + 2k$$

where k is the number of parameters.

Goodness of Fit

The deviance measures how well our model fits:

$$D = -2 \ln \mathcal{L}(\hat{\theta})$$

Richardson-Lucy Deconvolution

Problem: We observe a distorted luminosity function [Richardson, 1972, Lucy, 1974]

$$\Phi_{obs}(m) = \int K(m, m')\Phi_{true}(m')dm'$$

Where $K(m, m')$ is the kernel representing

- Completeness: stars at magnitude m' might not be detected
- Photometric errors: star with true magnitude m' measured as m

$$K(m, m') = C(m') \cdot P(m|m')$$

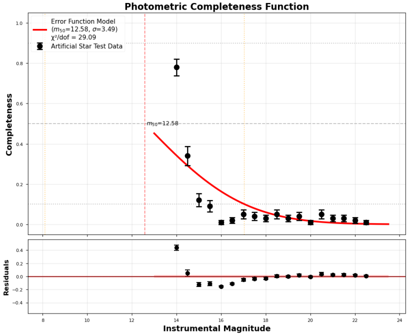
assuming Gaussian photometric errors

$$P(m|m') = \frac{1}{\sqrt{2\pi\sigma_m^2}} \exp\left[-\frac{(m - m')^2}{2\sigma_m^2}\right]$$

Richardson-Lucy Algorithm

We want to recover Φ_{true} from Φ_{obs}. The naive approach is $\Phi_{true} = \Phi_{obs}/C(m)$. This ignores scattering. Instead we aim to maximize the likelihood under Poisson statistics and enforce positivity. Start with a guess $\Phi_{true}^{(0)} = \Phi_{obs}$ and iterate

$$\Phi_{true}^{(n+1)}(m') = \Phi_{true}^{(n)}(m') \int \frac{\Phi_{obs}(m)}{\left[\int K(m, m'')\Phi_{true}^{(n)}(m'')dm''\right]} K(m, m')dm$$



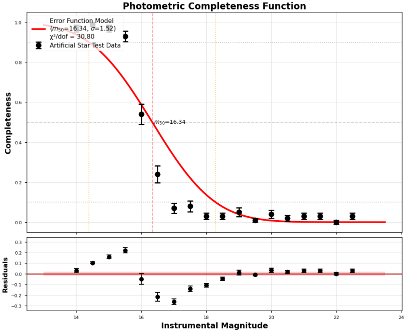


Figure: Completeness functions for M2 (top) and M34 (bottom)

The Fundemental Problem

We do not just observe stars in the cluster
$N_{observed}(m, r, \mu, \text{color}) = N_{cluster} + N_{foreground} + N_{background}$ **Goal**: For each detected star i, assign a membership probability $P_{member,i} \in [0, 1]$.
Define

$$\text{Purity} = \frac{N_{\text{true members classified as members}}}{N_{\text{all stars classified as members}}}$$

$$\text{Completeness} = \frac{N_{\text{true members classified as members}}}{N_{\text{all true members}}}$$

$$F_1 = 2 \cdot \frac{\text{Purity} \times \text{Completeness}}{\text{Purity} + \text{Completeness}}$$

CMD Filtering: Theory

Stars in a cluster [Bressan et al., 2012, Cardelli et al., 1989]

- Same age $\rightarrow$ follow single isochrone
- Same metallicity $[Fe/H]$
- Same distance $\rightarrow$ single distance modulus $\mu = m - M$
- Same reddening $E(B - V)$

The observed CMD:

$$(g - r, g) \approx (\text{isochrone}_{[Fe/H],age}(M_g) + E(g - r), M_g + \mu + A_g)$$

Isochrone Distance

The perpendicular distance from a star to the isochrone:

$$d_{CMD}(i) = \min_j \sqrt{\left(\frac{g_i - g_{iso,j}}{\sigma_g}\right)^2 + \left(\frac{(g-r)_i - (g-r)_{iso,j}}{\sigma_{g-r}}\right)^2}$$

Gaussian probability

$$P_{CMD}(i) = \exp\left[-\frac{d^2_{CMD}(i)}{2}\right]$$

and photometric error

$$P(c_i, m_i | c_{true}, m_{true}) = \mathcal{N}\left(\begin{bmatrix} c_{true} \\ m_{true} \end{bmatrix}, \begin{bmatrix} \sigma_c^2 & \rho\sigma_c\sigma_m \\ \rho\sigma_c\sigma_m & \sigma_m^2 \end{bmatrix}\right)$$

marginalized probability

$$P_{CMD}(i) = \int P(\text{on isochrone}|c_{true}, m_{true}) \cdot P(c_i, m_i | c_{true}, m_{true}) dc_{true} dm_{true}$$

Proper Motion Distribution

Cluster stars move coherently, field stars move randomly. Follow a bivariate Gaussian in proper motion-space [Gaia Collaboration et al., 2023]

$$P(\mu_i|\text{cluster}) = \mathcal{N}(\mu_i; \mu_{cluster}, \Sigma_{cluster})$$

where:

$$\mu_{cluster} = \begin{bmatrix} \overline{\mu_\alpha \cos\delta} \\ \overline{\mu_\delta} \end{bmatrix}, \qquad \Sigma_{cluster} = \begin{bmatrix} \sigma^2_{\mu_\alpha} & \rho\sigma_{\mu_\alpha}\sigma_{\mu_\delta} \\ \rho\sigma_{\mu_\alpha}\sigma_{\mu_\delta} & \sigma^2_{\mu_\delta} \end{bmatrix}$$

Proper Motion Membership Probability

Mahalanobis distance:

$$D_i^2 = (\mu_i - \mu_{cluster})^T \Sigma_{cluster}^{-1} (\mu_i - \mu_{cluster})$$

we also have uncertainty in the observed PM. The total covariance is

$$\Sigma_{total,i} = \Sigma_{cluster} + \Sigma_{obs,i} \quad \text{where} \quad \Sigma_{obs,i} = \begin{bmatrix} \sigma^2_{\mu_\alpha,i} & 0 \\ 0 & \sigma^2_{\mu_\delta,i} \end{bmatrix}$$

Membership Probability:

$$P_{PM}(i) = \frac{\mathcal{L}_{cluster}(i)}{\mathcal{L}_{cluster}(i) + \mathcal{L}_{field}(i)}$$

where (and similarly for field)

$$\mathcal{L}_{cluster}(i) = \frac{1}{2\pi |\Sigma_{total,i}|^{1/2}} \exp\left[-\frac{1}{2}(\mu_i - \mu_{cluster})^T \Sigma_{total,i}^{-1} (\mu_i - \mu_{cluster})\right]$$

Radial Profile Method

Cluster stars are centrally concentrated, field stars are uniformly distributed. Surface density:

$$\Sigma(r) = \Sigma_{cluster}(r) + \Sigma_{background}$$

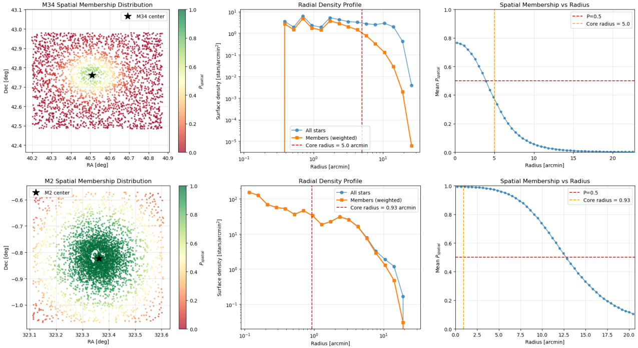


Figure: Spatial likelihood based on radial distance from cluster center. Top: M34, Bottom: M2

The Challenge with Mass Estimation

To measure a mass density profile $\rho_{mass}(r)$, not just number density $n(r)$.
We use [Kroupa, 2001, Henry and McCarthy, 2004]

$$\rho_{mass}(r) = \int_0^\infty \phi(M, r) \cdot M dM$$

where $\phi(M, r)$ is the mass function.
Problem: We observe light, not mass
The Observable Chain:
Detector counts $\rightarrow$ Apparent magnitude m $\rightarrow$ Absolute magnitude M $\rightarrow$ Luminosity L $\rightarrow$ Mass $\mathcal{M}$

Case Study: M2

- Located in the constellation Aquarius.
- Extremely old, estimated at 12.5 to 13 billion years.
- Very large, with a diameter of 150 ly.

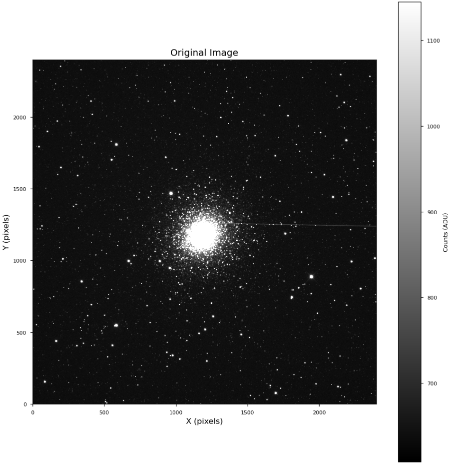


Figure: *

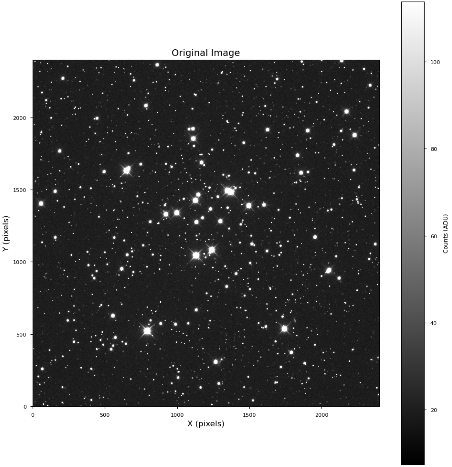
The globular cluster M2

M2's Mass Density Profile

- Characterized by a highly concentrated core.
- High central density, with a very steep profile.
- Well-modeled by standard globular cluster profiles (e.g., King or Wilson models).
- High gravitational binding results in a very stable, spherically symmetric profile.

Case Study: M34

- Located in the constellation Perseus.
- Intermediate age, around 200 million years.
- Smaller and less populous, with 100 to 400 members.



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The open cluster M34

M34's Mass Density Profile

- Much lower overall stellar density compared to M2.
- Less concentrated core and a much shallower profile.
- Not as symmetrical, reflecting its "fluffier" and less bound nature.
- Susceptible to tidal forces from the Milky Way, leading to its eventual dispersion.

M2 vs. M34: A Direct Comparison

Feature	**Globular Cluster M2**	**Open Cluster M34**
Age	$\sim$13 billion years	$\sim$200 million years
Population	$> 150,000$ stars	100–400 stars
Shape	Highly spherical, compact	Irregular, loosely packed
Density Profile	Steep, highly peaked	Shallow, less concentrated
Stability	Dynamically stable	Prone to disruption

Table: Comparison of key features for M2 and M34.

Analysis Overview

- Use the **cleaned member catalog** to compare the internal structure of M2 and M34.
- Focus on three main diagnostics:
    - Color–magnitude diagrams (CMDs) for stellar populations.
    - Radial distribution of stellar mass (mass segregation).
    - Line-of-sight velocity dispersions and relaxation times.
- Goal: determine how far each cluster has evolved dynamically and how strongly mass is segregated.

Color–Magnitude Diagrams: M34 vs. M2

M34 (open cluster)

- Young, metal-rich main sequence with a clear turnoff.
- Only a small number of evolved stars.
- CMD consistent with an age of $\sim 2 \times 10^8$ yr at $d \approx 470\,\mathrm{pc}$.

M2 (globular cluster)

- Old, metal-poor population dominated by red giants.
- Extended horizontal branch and giant branch.
- CMD consistent with an age of $\sim 1.2 \times 10^{10}$ yr at $d \approx 11.5\,\mathrm{kpc}$.

CMD plots

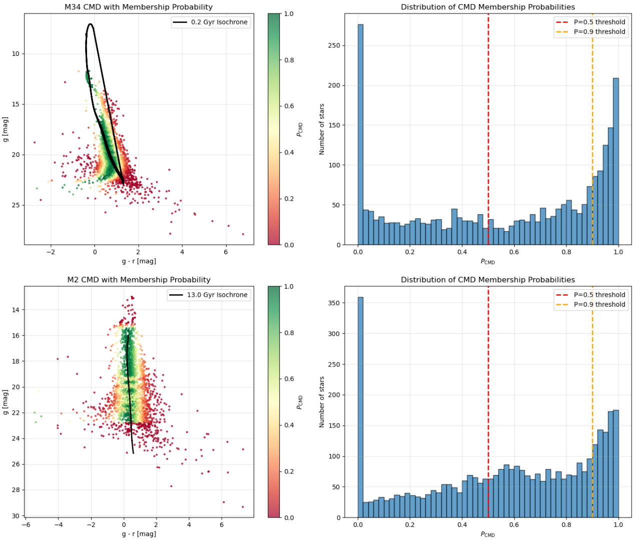

Figure: CMDs for M34 (left) and M2 (right)

Radial Distribution by Stellar Mass

- Use r-band magnitude as a proxy for stellar mass.
- For each cluster:
  - Select high-probability members ($P_{\rm mem} \gtrsim 0.8$).
  - Rank stars by r and define:
    - "Bright" sample $=$ brightest 30% (higher-mass stars).
    - "Faint" sample $=$ faintest 30% (lower-mass stars).
  - Compare cumulative radial distributions $F_{\rm bright}(R)$ and $F_{\rm faint}(R)$.
- Quantify the difference with a two-sample Kolmogorov–Smirnov (K–S) test.

Mass Segregation: K–S Test Results

Cluster	D	p-value	Interpretation
M34	0.126	1.1×10^{-5}	Moderate, significant segregation
M2	0.400	1.2×10^{-84}	Very strong segregation

- In both clusters, bright (high-mass) stars are more centrally concentrated than faint stars.
- Effect is **much stronger** in M2, consistent with its status as an old, dense globular cluster.
- M34 shows clear but only moderate segregation, typical of a dynamically evolved open cluster.

Synthetic Velocity Dispersion Profiles

- Direct Gaia DR3 radial velocities could not be accessed reliably, so we constructed **synthetic** Gaia-like radial velocities:
  - For each member star, draw $v_{r,i} \sim \mathcal{N}(v_0, \sigma_v(R_i))$.
  - Adopt nearly constant $\sigma_v \approx 0.5\ \mathrm{km\,s^{-1}}$ for M34.
  - For M2 use a declining profile: high central σ_v that falls toward the outskirts.
- Bin stars in radius and estimate the intrinsic dispersion in each bin using

$$\sigma_v^2 = \frac{1}{N-1}\sum_i (v_{r,i} - \bar{v}_r)^2 - \frac{1}{N}\sum_i \epsilon_i^2,$$

  where ϵ_i are measurement errors.
- Recovered global dispersions:
  - M34: $\sigma_v \approx 0.5 \pm 0.1\ \mathrm{km\,s^{-1}}$ (cold, nearly flat profile).
  - M2: $\sigma_v \approx 10 \pm 1\ \mathrm{km\,s^{-1}}$ (hot center, declining with radius).

Relaxation Time and Dynamical State

- Two-body relaxation time at the half-mass radius:

$$t_{\rm relax} \approx \frac{0.1N}{\ln N}\, t_{\rm cross}, \qquad t_{\rm cross} = \frac{R_h}{\sigma_v}.$$

- Using synthetic dispersions and structural parameters:

Cluster	N	R_h [pc]	$t_{\rm relax}$	$t_{\rm age}/t_{\rm relax}$
M34	~ 400	0.30	~ 4 Myr	~ 50
M2	$\sim 10^5$	10	~ 0.70 Gyr	~ 17

- Both clusters are older than their relaxation times, so significant mass segregation is expected.
- M2 has experienced many relaxation cycles over a Hubble time; M34, though young, is also dynamically evolved because $t_{\rm relax}$ is very short.

[Binney and Tremaine, 2008]

Dynamical Interpretation

- **M2**:
  - Strong mass segregation ($D = 0.4$) and high central σ_v.
  - Long history of many relaxation times $\Rightarrow$ dynamically old, partially equipartitioned globular cluster.
- **M34**:
  - Moderate but significant mass segregation.
  - Very short $t_{\rm relax}$ relative to its age $\Rightarrow$ dynamically evolved despite being young in absolute terms.
- Neither cluster reaches full energy equipartition; simulations predict only *partial* equipartition even for very old clusters.
- Together, M2 and M34 illustrate how cluster mass, density, and age shape the observed degree of mass segregation and internal kinematics.

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Thank You